

Week-1 SQL Commands

By Shaaz

Table Creation

Client Table Master

```
CREATE TABLE Client_Master (  
    ClientNo INT IDENTITY(1,1) PRIMARY KEY,  
    Name VARCHAR(25) NOT NULL,  
    Address1 VARCHAR(30),  
    Address2 VARCHAR(30),  
    City VARCHAR(15),  
    Pincode INT,  
    State VARCHAR(15),  
    BalDue DECIMAL(10,2)  
);
```

Product Table Master

```
CREATE TABLE Product_Master (  
    ProductNo INT IDENTITY(1,1) PRIMARY KEY,  
    Description VARCHAR(25) NOT NULL,  
    ProfitPercent DECIMAL(5,2) NOT NULL,  
    UnitMeasure VARCHAR(10) NOT NULL,  
    QtyOnHand INT NOT NULL,  
    ReorderLvl INT NOT NULL,  
    SellPrice DECIMAL(8,2) NOT NULL CHECK (SellPrice > 0),  
    CostPrice DECIMAL(8,2) NOT NULL CHECK (CostPrice > 0)  
);
```

Sales Table Master

```
CREATE TABLE Salesman_Master (  
    SalesmanNo INT IDENTITY(1,1) PRIMARY KEY,  
    SalesmanName VARCHAR(25) NOT NULL,  
    Address1 VARCHAR(30),  
    Address2 VARCHAR(30),  
    City VARCHAR(20),
```

```

Pincode INT,
State VARCHAR(20),
Target DECIMAL(8,2) NOT NULL CHECK (Target > 0),
YTDSales DECIMAL(8,2) NOT NULL,
Remarks VARCHAR(60)
);

```

Sales Order Table

```

CREATE TABLE Sales_Order (
    OrderNo INT IDENTITY(1,1) PRIMARY KEY,
    OrderDate DATE NOT NULL,
    ClientNo INT NOT NULL,
    SalesmanNo INT NOT NULL,
    DeliveryType CHAR(1) CHECK (DeliveryType IN ('P','F')),
    BilledYN CHAR(1) CHECK (BilledYN IN ('Y','N')),
    DeliveryDate DATE,
    OrderStatus VARCHAR(10) CHECK (OrderStatus IN ('In
Process','Fulfilled','Backorder','Cancelled')),

    CONSTRAINT FK_SalesOrder_Client
        FOREIGN KEY (ClientNo) REFERENCES Client_Master(ClientNo),

    CONSTRAINT FK_SalesOrder_Salesman
        FOREIGN KEY (SalesmanNo) REFERENCES Salesman_Master(SalesmanNo)
);

```

Sales order details Table

```

CREATE TABLE Sales_Order_Details (
    OrderDetailId INT IDENTITY(1,1) PRIMARY KEY,
    OrderNo INT NOT NULL,
    ProductNo INT NOT NULL,
    QtyOrdered INT NOT NULL CHECK (QtyOrdered > 0),
    QtyDisp INT NOT NULL CHECK (QtyDisp >= 0),
    ProductRate DECIMAL(8,2) NOT NULL,

    CONSTRAINT FK_OrderDetails_Order
        FOREIGN KEY (OrderNo) REFERENCES Sales_Order(OrderNo),

    CONSTRAINT FK_OrderDetails_Product
        FOREIGN KEY (ProductNo) REFERENCES Product_Master(ProductNo)
);

```

Insertions into Table:-

```
INSERT INTO Client_Master (Name, Address1, City, Pincode, State, BalDue)
VALUES
```

```
('Ivan Bayross', 'Andheri East', 'Mumbai', 400054, 'Maharashtra', 15000),
('Sunil Shah', 'Dadar', 'Mumbai', 400028, 'Maharashtra', 4500),
('Ravi Kumar', 'Banjara Hills', 'Hyderabad', 500034, 'Telangana', 7000),
('Anita Verma', 'Indiranagar', 'Bangalore', 560038, 'Karnataka', 2500),
('John Mathew', 'Fort Kochi', 'Kochi', 682001, 'Kerala', 18000);
```

```
INSERT INTO Product_Master
```

```
(Description, ProfitPercent, UnitMeasure, QtyOnHand, ReorderLvl, SellPrice,
CostPrice)
```

```
VALUES
```

```
('Shirts', 10, 'Piece', 200, 50, 3000, 2500),
('Jeans', 12, 'Piece', 150, 40, 4500, 4000),
('Jackets', 15, 'Piece', 50, 20, 5000, 4500),
('Trousers', 8, 'Piece', 0, 30, 2800, 2300), -- non-moving
('T-Shirts', 5, 'Piece', 500, 100, 1200, 900);
```

```
INSERT INTO Salesman_Master
```

```
(SalesmanName, Address1, City, Pincode, State, Target, YTDSales, Remarks)
```

```
VALUES
```

```
('Aman', 'Malad', 'Mumbai', 400064, 'Maharashtra', 100, 50, 'Good'),
('Rohit', 'Andheri', 'Mumbai', 400053, 'Maharashtra', 20000, 15000, 'Excellent'),
('Suresh', 'Secunderabad', 'Hyderabad', 500003, 'Telangana', 12000, 6000,
'Average'),
('Anil', 'Whitefield', 'Bangalore', 560066, 'Karnataka', 9000, 3000, 'New'),
('Kiran', 'Vytila', 'Kochi', 682019, 'Kerala', 18000, 12000, 'Senior');
```

```
INSERT INTO Sales_Order
```

```
(OrderDate, ClientNo, SalesmanNo, DeliveryType, BilledYN, DeliveryDate,
OrderStatus)
```

```
VALUES
```

```
('2002-04-20', 1, 1, 'P', 'Y', '2002-04-25', 'Fulfilled'),
('2002-05-10', 2, 2, 'F', 'Y', '2002-05-15', 'Fulfilled'),
('2002-05-18', 3, 3, 'P', 'N', NULL, 'In Process'),
('2002-06-05', 4, 4, 'F', 'N', NULL, 'Backorder'),
('2002-05-22', 1, 1, 'P', 'Y', '2002-05-30', 'Fulfilled');
```

```

INSERT INTO Sales_Order_Details
(OrderNo, ProductNo, QtyOrdered, QtyDisp, ProductRate)
VALUES
(1, 1, 10, 10, 3000),
(2, 2, 3, 3, 4500),      -- < 5 units
(3, 3, 6, 0, 5000),
(4, 4, 4, 0, 2800),      -- non-moving
(5, 5, 20, 20, 1200);

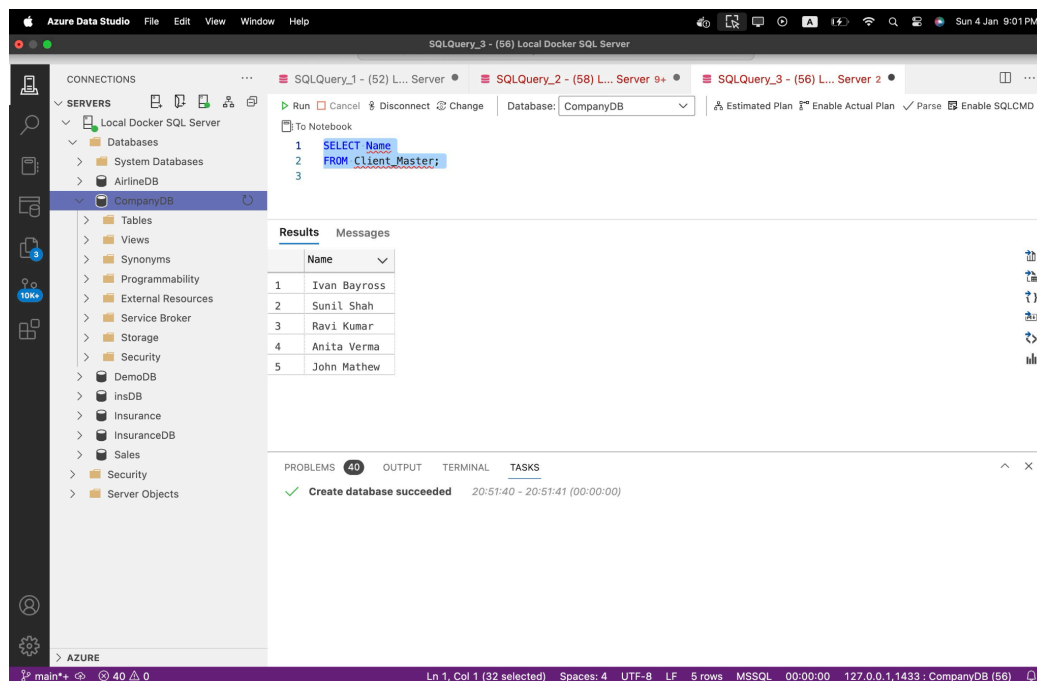
```

1. Display the names of all the clients

```

SELECT Name
FROM Client_Master;

```



2. Display all the clients who are located in Mumbai

```
SELECT *  
FROM Client_Master  
WHERE City = 'Mumbai';
```

The screenshot shows the Azure Data Studio interface. On the left, the 'SERVERS' pane shows a connection to 'Local Docker SQL Server' with a database named 'CompanyDB'. The main editor displays a SQL query: `SELECT * FROM Client_Master WHERE City = 'Mumbai';`. The 'Results' pane shows the output of the query, which is a table with 10 columns: ClientNo, Name, Address1, Address2, City, Pincode, State, BalDue, and an unnamed column. The results are as follows:

ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue	
1	Ivan Bayross	Andheri East	NULL	Mumbai	400054	Maharashtra	15000.00	
2	Sunil Shah	Dadar	NULL	Mumbai	400028	Maharashtra	4500.00	

The bottom status bar indicates the query was executed successfully on 'CompanyDB (56)'.

3. Display all the products whose selling price is > 2000 and < 5000

```
SELECT *  
FROM Product_Master  
WHERE SellPrice > 2000  
AND SellPrice < 5000;
```

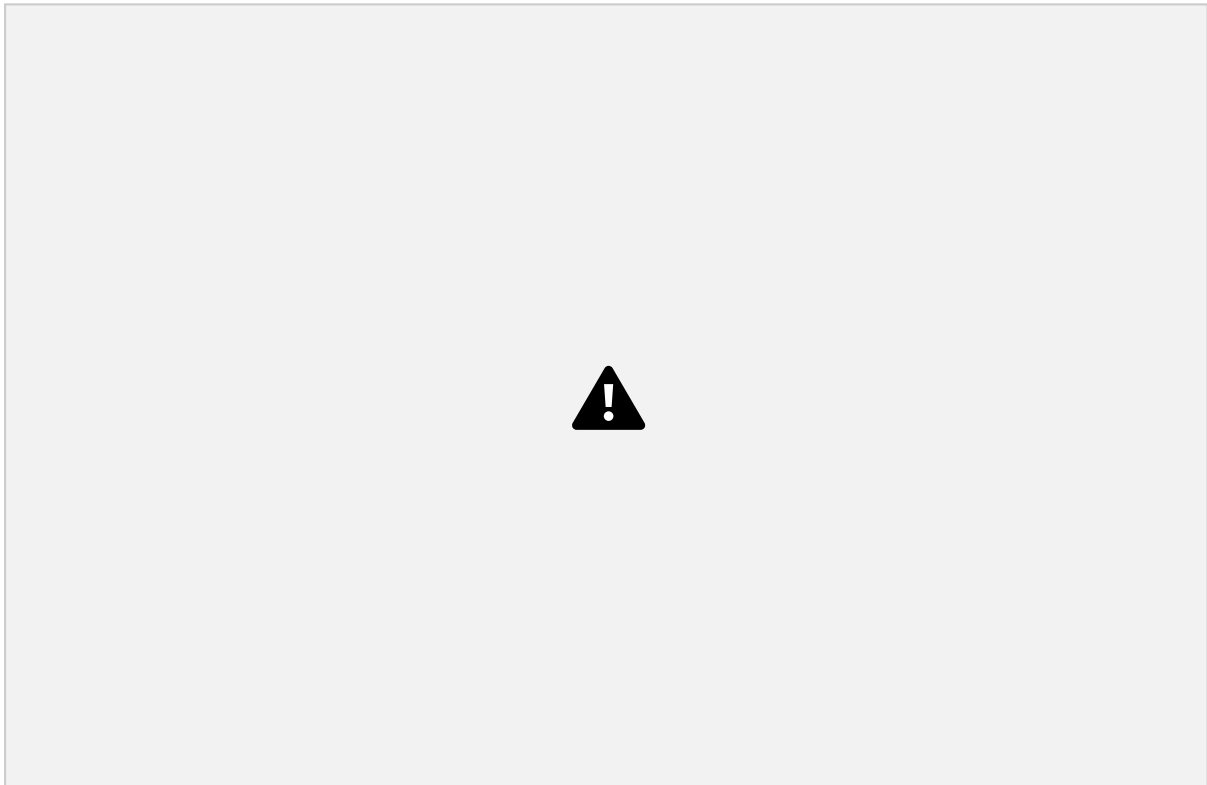
The screenshot shows the Azure Data Studio interface. On the left, the 'SERVERS' pane shows a connection to 'Local Docker SQL Server' with a database named 'CompanyDB'. The main editor displays a SQL query: `SELECT * FROM Product_Master WHERE SellPrice > 2000 AND SellPrice < 5000;`. The 'Results' pane shows the output of the query, which is a table with 10 columns: ProductNo, Description, ProfitPercent, UnitMeasure, QtyOnHand, ReorderLvl, SellPrice, and two unnamed columns. The results are as follows:

ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice		
1	Shirts	10.00	Piece	200	50	3000.00	2	
2	Jeans	12.00	Piece	150	40	4500.00	4	
3	Trousers	8.00	Piece	0	30	2800.00	2	

The bottom status bar indicates the query was executed successfully on 'CompanyDB (56)'.

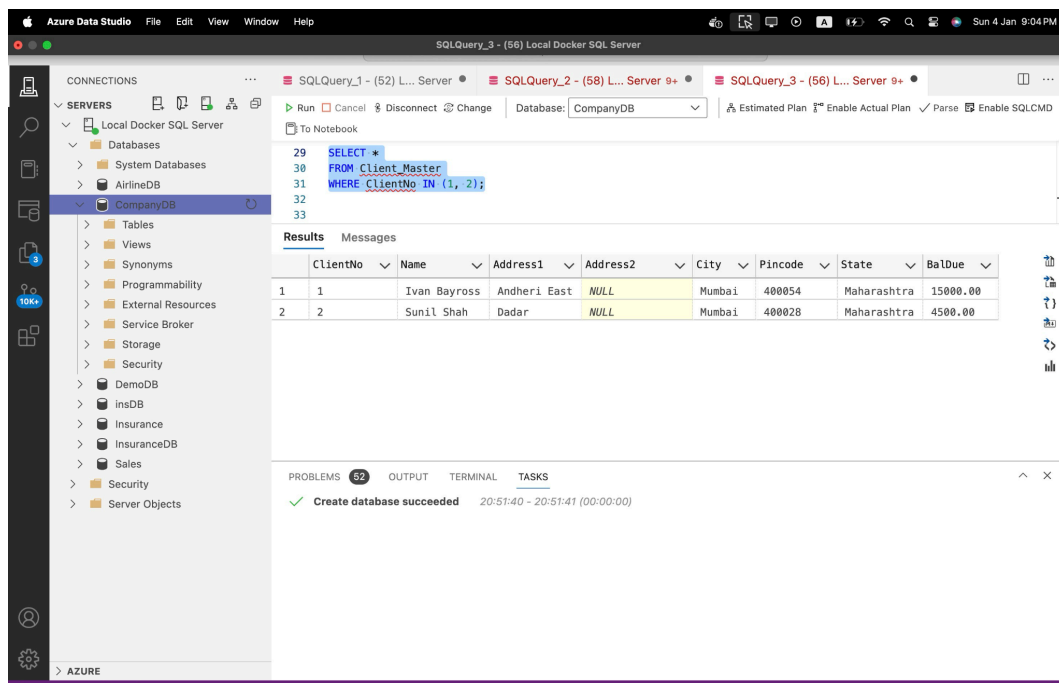
4. Display Name, City and State of Clients not in the state of Maharashtra

```
SELECT Name, City, State
FROM Client_Master
WHERE State <> 'Maharashtra';
```



5. Display all the information of client no C0001 and C0002

```
SELECT *
FROM Client_Master
WHERE ClientNo IN (1, 2);
```



The screenshot shows the Azure Data Studio interface. The left sidebar displays the 'CONNECTIONS' pane with a tree view of the database structure, including 'Local Docker SQL Server', 'System Databases', 'AirlineDB', and 'CompanyDB'. The main editor window shows a SQL query in the 'SQLQuery_3 - (56) Local Docker SQL Server' tab. The query is:

```
29 SELECT *
30 FROM Client_Master
31 WHERE ClientNo IN (1, 2);
32
33
```

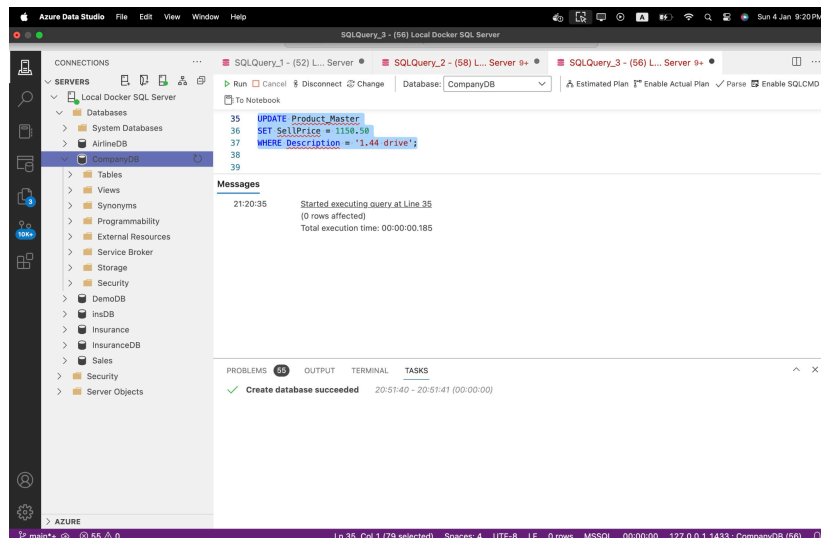
The 'Results' pane shows the output of the query, displaying a table with 10 columns: ClientNo, Name, Address1, Address2, City, Pincode, State, and BalDue. The results are as follows:

ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	Ivan Bayross	Andheri East	NULL	Mumbai	400054	Maharashtra	15000.00
2	Sunil Shah	Dadar	NULL	Mumbai	400028	Maharashtra	4500.00

The bottom status bar shows a message: 'Create database succeeded' with a green checkmark icon.

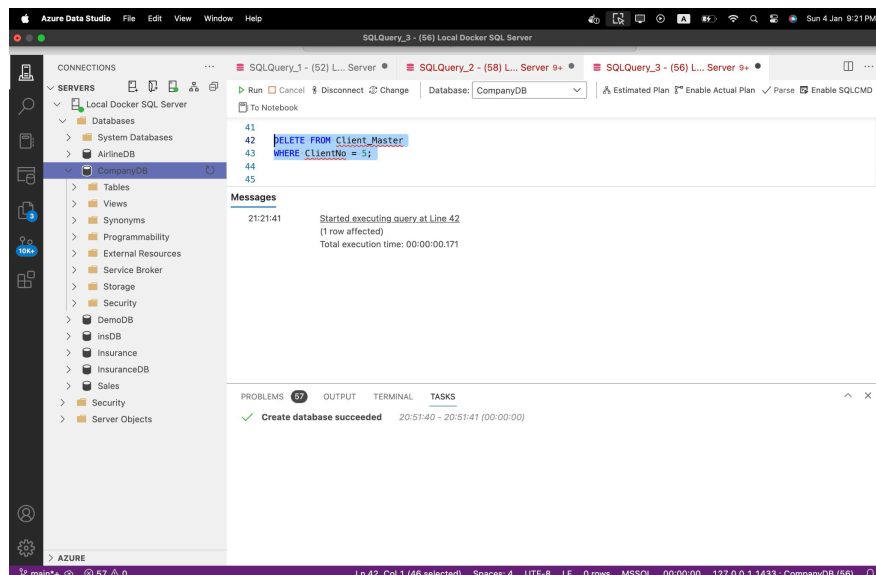
6. Change the selling price of '1.44 drive' to Rs. 1150.50

```
UPDATE Product_Master  
SET SellPrice = 1150.50  
WHERE Description = '1.44 drive';
```



7. Delete the record of client_no C0005

```
DELETE FROM Client_Master  
WHERE ClientNo = 5;
```



8. Display the clients who stay in a city whose second letter is 'a'

```
SELECT *  
FROM Client_Master  
WHERE City LIKE '_a%';
```

The screenshot shows the Azure Data Studio interface. On the left, the 'CONNECTIONS' pane shows a tree view of servers and databases. The 'CompanyDB' database is selected. The main editor shows a SQL query:

```
47 SELECT *
48 FROM Client_Master
49 WHERE City LIKE '_a%';
50
51
```

The 'Results' pane shows a table with the following data:

	ClientNo	Name	Address1	Address2	City	Pincode	State	BalDue
1	4	Anita Verma	Indiranagar	NULL	Bangalore	560038	Karnataka	2500.00

The 'PROBLEMS' pane shows a message: 'Create database succeeded'.

9. Count the number of products having price ≥ 1500

```
SELECT COUNT(*) AS Product_Count
FROM Product_Master
WHERE SellPrice >= 1500;
```

The screenshot shows the Azure Data Studio interface. On the left, the 'CONNECTIONS' pane shows a tree view of servers and databases. The 'CompanyDB' database is selected. The main editor shows a SQL query:

```
53 SELECT COUNT(*) AS Product_Count
54 FROM Product_Master
55 WHERE SellPrice >= 1500;
56
57
```

The 'Results' pane shows a table with the following data:

	Product_Count
1	4

The 'PROBLEMS' pane shows a message: 'Create database succeeded'.

10. Display qtyordered, qtydisp and balancedqty

SELECT

QtyOrdered,

QtyDisp,

(QtyOrdered - QtyDisp) AS BalancedQty

FROM Sales_Order_Details;

The screenshot shows the Azure Data Studio interface. The left sidebar displays the 'CONNECTIONS' pane with a tree view of the 'Local Docker SQL Server' and 'CompanyDB' databases. The main editor window shows a SQL query being executed. The query is as follows:

```
57
58
59 SELECT
60     QtyOrdered,
61     QtyDisp,
62     (QtyOrdered - QtyDisp) AS BalancedQty
63 FROM Sales_Order_Details;
64
```

Below the query editor, the 'Results' pane displays the output of the query. The results are shown in a table with the following columns: QtyOrdered, QtyDisp, and BalancedQty. The table contains 5 rows of data.

	QtyOrdered	QtyDisp	BalancedQty
1	10	10	0
2	3	3	0
3	6	0	6
4	4	0	4
5	20	20	0

The bottom status bar shows the current file is 'main+', the cursor is at line 59, column 1 (104 selected), and the database is 'CompanyDB (56)'.

Write Commands to do following:-

1. Make ClientNo as primary key in Client_Master

```
ALTER TABLE Client_Master  
ADD CONSTRAINT PK_ClientMaster PRIMARY KEY (ClientNo);
```

2. Add a new column Phone_No in the Client_Master table

```
ALTER TABLE Client_Master  
ADD Phone_No VARCHAR(15);
```

3. Add NOT NULL constraint in Product_Master table for

Description, ProfitPercent, SellPrice, and CostPrice:-

```
ALTER TABLE Product_Master
```

```
ALTER COLUMN Description VARCHAR(25) NOT NULL;
```

```
ALTER TABLE Product_Master
```

```
ALTER COLUMN ProfitPercent DECIMAL(5,2) NOT NULL;
```

```
ALTER TABLE Product_Master
```

```
ALTER COLUMN SellPrice DECIMAL(8,2) NOT NULL;
```

```
ALTER TABLE Product_Master
```

```
ALTER COLUMN CostPrice DECIMAL(8,2) NOT NULL;
```

4. Change the size of Name column to 60 in Client_Master

```
ALTER TABLE Client_Master
```

```
ALTER COLUMN Name VARCHAR(60);
```

5. Remove Pincod column from Client_Master

```
ALTER TABLE Client_Master
```

```
DROP COLUMN Pincod;
```

Write queries for following descriptions (JOINS):-

1. Find out the products which have been sold to 'Ivan Bayross'

```
SELECT DISTINCT p.Description
```

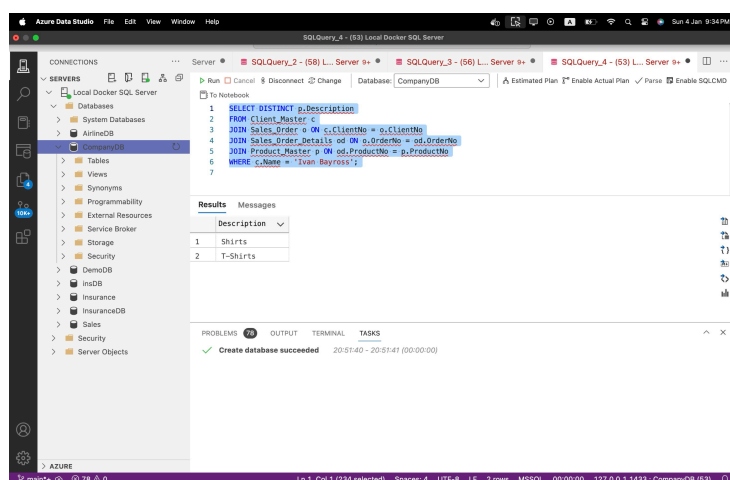
```
FROM Client_Master c
```

```
JOIN Sales_Order o ON c.ClientNo = o.ClientNo
```

```
JOIN Sales_Order_Details od ON o.OrderNo = od.OrderNo
```

```
JOIN Product_Master p ON od.ProductNo = p.ProductNo
```

```
WHERE c.Name = 'Ivan Bayross';
```



2. Find the products and their quantities that will have to be delivered in the **current month**

```
SELECT p.Description, od.QtyOrdered
FROM Sales_Order o
JOIN Sales_Order_Details od ON o.OrderNo = od.OrderNo
JOIN Product_Master p ON od.ProductNo = p.ProductNo
WHERE MONTH(o.DeliveryDate) = MONTH(GETDATE())
AND YEAR(o.DeliveryDate) = YEAR(GETDATE());
```

The screenshot shows the Azure Data Studio interface. On the left, the 'CONNECTIONS' pane shows a tree view of the 'Local Docker SQL Server' with 'CompanyDB' selected. The main editor displays a SQL query. Below the query, the 'Results' pane shows a table with two columns: 'Description' and 'QtyOrdered'. The results table contains one row: 'Pull Overs' with a quantity of 3. The bottom status bar shows 'Ln 1, Col 1 (254 selected) Spaces: 4 UTF-8 LF 1 rows MSSQL 00:00:00 127.0.0.1,1433 : CompanyDB (57)'.

Description	QtyOrdered
Pull Overs	3

3. List the **ProductNo** and **Description** of constantly sold

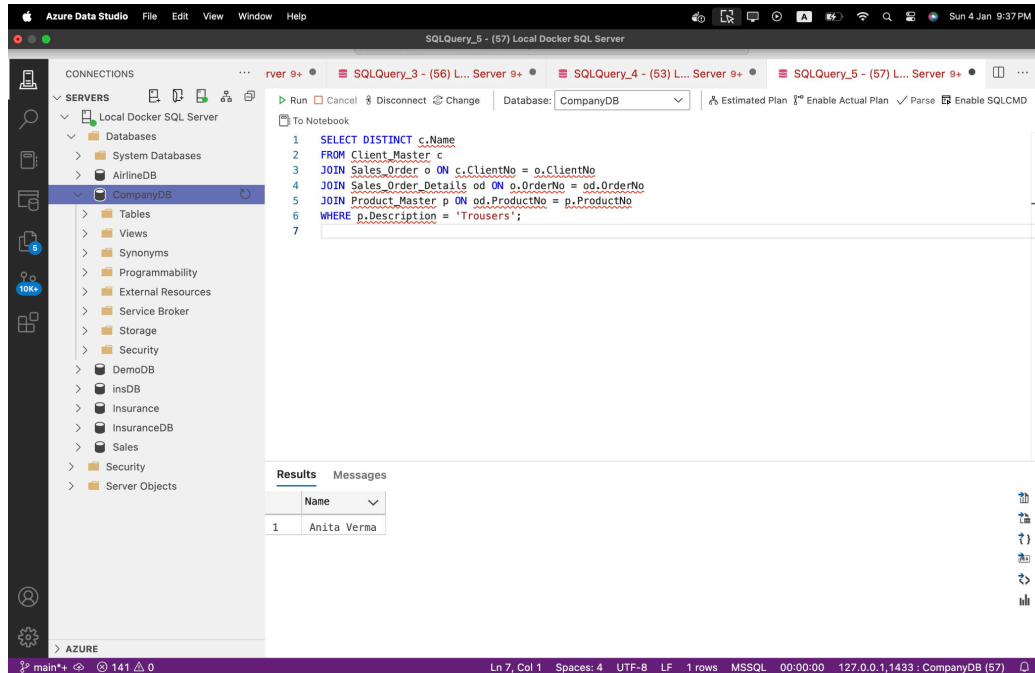
```
SELECT p.ProductNo, p.Description
FROM Product_Master p
JOIN Sales_Order_Details od ON p.ProductNo = od.ProductNo
GROUP BY p.ProductNo, p.Description
HAVING COUNT(od.ProductNo) > 1;
```

The screenshot shows the Azure Data Studio interface. On the left, the 'CONNECTIONS' pane shows a tree view of the 'Local Docker SQL Server' with 'CompanyDB' selected. The main editor displays a SQL query. Below the query, the 'Results' pane shows a table with two columns: 'ProductNo' and 'Description'. The results table contains one row: '6' with the description 'Pull Overs'. The bottom status bar shows 'Ln 1, Col 1 (254 selected) Spaces: 4 UTF-8 LF 1 rows MSSQL 00:00:00 127.0.0.1,1433 : CompanyDB (57)'.

ProductNo	Description
6	Pull Overs

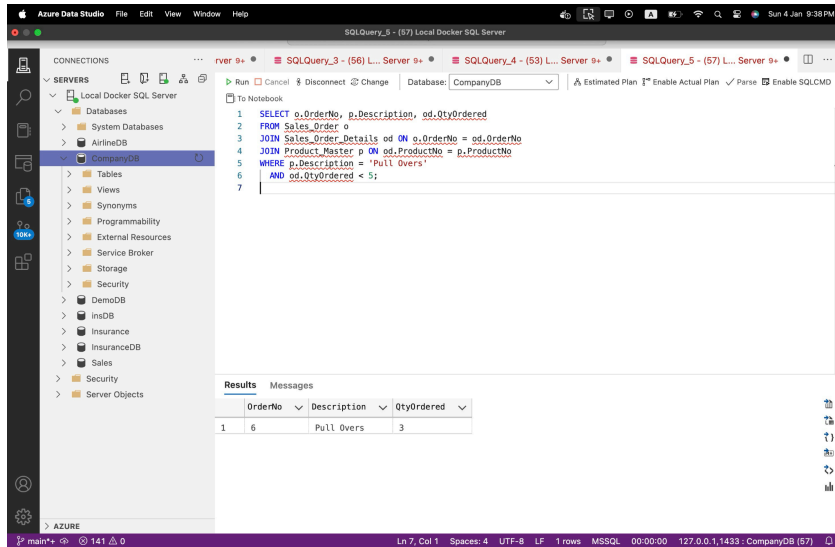
4. Find the names of clients who have purchased 'Trousers'

```
SELECT DISTINCT c.Name
FROM Client_Master c
JOIN Sales_Order o ON c.ClientNo = o.ClientNo
JOIN Sales_Order_Details od ON o.OrderNo = od.OrderNo
JOIN Product_Master p ON od.ProductNo = p.ProductNo
WHERE p.Description = 'Trousers';
```



5. List the products and orders from customers who have ordered

```
SELECT o.OrderNo, p.Description, od.QtyOrdered
FROM Sales_Order o
JOIN Sales_Order_Details od ON o.OrderNo = od.OrderNo
JOIN Product_Master p ON od.ProductNo = p.ProductNo
WHERE p.Description = 'Pull Overs'
AND od.QtyOrdered < 5;
```



Write queries for following descriptions: (Subqueries):-

1. Finding the non-moving products:-

```

SELECT *
FROM Product_Master
WHERE ProductNo NOT IN (
    SELECT DISTINCT ProductNo
    FROM Sales_Order_Details
);

```

The screenshot shows the Azure Data Studio interface. The left sidebar displays the 'SERVERS' tree with 'Local Docker SQL Server' expanded, showing 'Databases' and 'CompanyDB'. The main editor shows a SQL query:

```

1 SELECT
2 FROM Product_Master
3 WHERE ProductNo NOT IN (
4     SELECT DISTINCT ProductNo
5     FROM Sales_Order_Details
6 )
7

```

The 'Results' pane at the bottom shows a table with 7 columns: ProductNo, Description, ProfitPercent, UnitMeasure, QtyOnHand, ReorderLvl, and SellPrice. The first row of data is:

ProductNo	Description	ProfitPercent	UnitMeasure	QtyOnHand	ReorderLvl	SellPrice
7	Trousers	10.00	Piece	80	15	3000.00

2. Finding the name and complete address for the customer

`SELECT Name, Address1, Address2, City, State`

`FROM Client_Master`

`WHERE ClientNo = (`

`SELECT ClientNo`

`FROM Sales_Order`

`WHERE OrderNo = '19001'`

`);`

The screenshot shows the Azure Data Studio interface. The left sidebar displays the 'SERVERS' tree with 'Local Docker SQL Server' expanded, showing 'Databases' and 'CompanyDB'. The main editor shows a SQL query:

```

1 SELECT Name, Address1, Address2, City, State
2 FROM Client_Master
3 WHERE ClientNo = (
4     SELECT ClientNo
5     FROM Sales_Order
6     WHERE OrderNo = '19001'
7 );
8
9
10
11

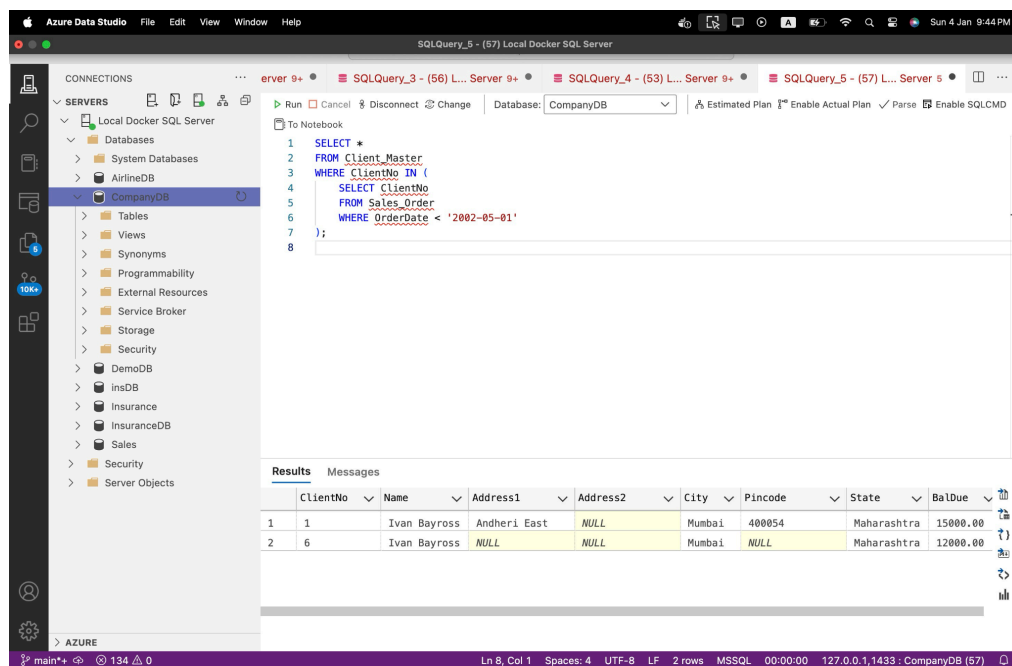
```

The 'Results' pane at the bottom shows a table with 5 columns: Name, Address1, Address2, City, and State. The first row of data is:

Name	Address1	Address2	City	State

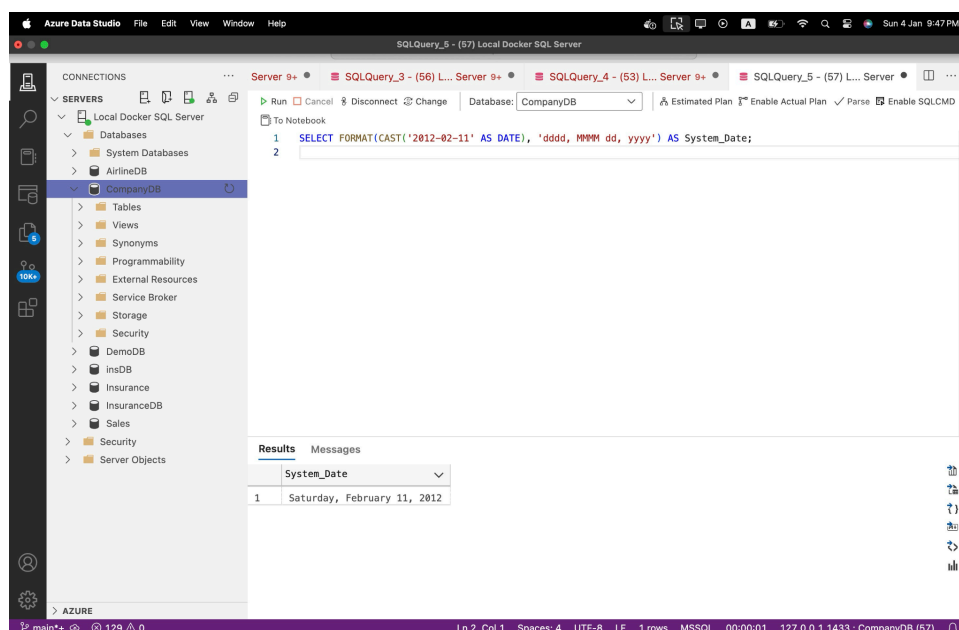
3. Finding the clients who have placed orders before the month of May'02

```
SELECT *  
FROM Client_Master  
WHERE ClientNo IN (  
    SELECT ClientNo  
    FROM Sales_Order  
    WHERE OrderDate < '2002-05-01'  
);
```



Write Commands to do following:-

1. Display system date as



2. Display **Balance Due** from Client master as

```
SELECT FORMAT(BalDue, '$99,999.99') AS Balance_Due
FROM Client_Master;
```

The screenshot shows the Azure Data Studio interface with a SQL query executed against the 'CompanyDB' database. The query is: `SELECT FORMAT(BalDue, '$99,999.99') AS Balance_Due FROM Client_Master;`. The results are displayed in a table with 7 rows, showing the formatted balance due for each client master entry.

	Balance_Due
1	\$999991500099
2	\$999994500099
3	\$999997000099
4	\$999992500099
5	\$999991200099
6	\$999998000099
7	\$999998000099

3. Display message as

"Salesman Aman sold goods of 50 while given target was 100."

```
SELECT
```

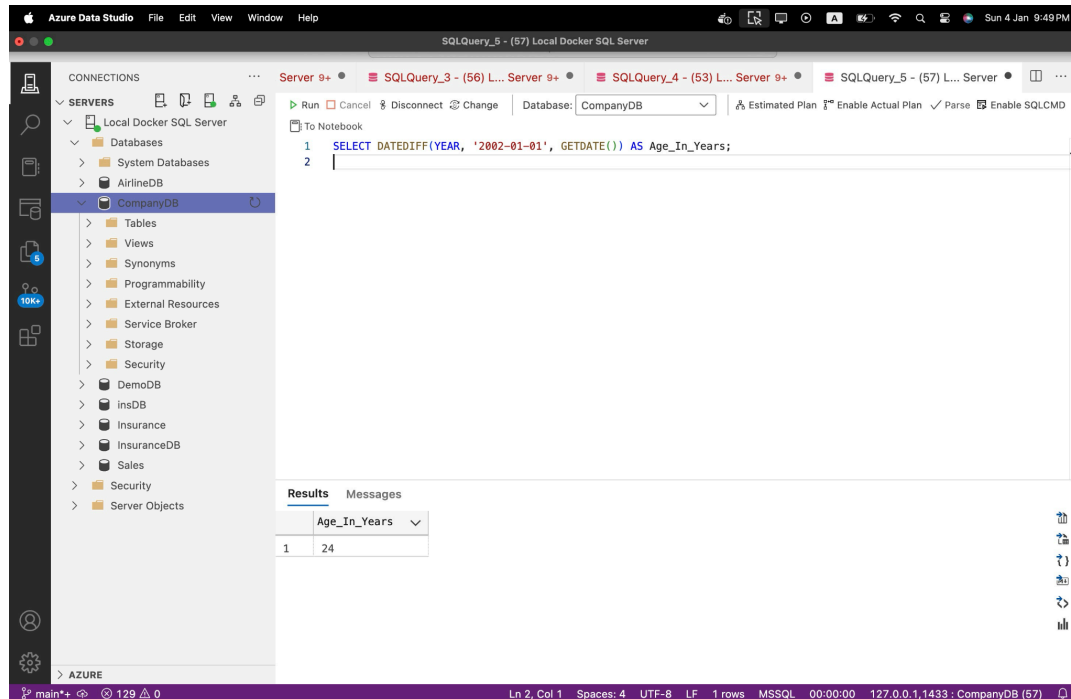
```
'Salesman Aman sold goods of 50 while given target was 100.' AS Message;
```

The screenshot shows the Azure Data Studio interface with a SQL query executed against the 'CompanyDB' database. The query is: `SELECT 'Salesman Aman sold goods of 50 while given target was 100.' AS Message;`. The results are displayed in a table with 1 row, showing the message.

	Message
1	Salesman Aman sold goods of 50 while given target was 100.

4. Display your **Age in Years**

```
SELECT DATEDIFF(YEAR, '2002-01-01', GETDATE()) AS Age_In_Years;
```



Define in 1 or 2 lines and give one example also:-

1. Recursive Relationship

Definition:

A recursive relationship is a relationship where a table is related to itself.

Example:

An **Employee** table where an employee reports to another employee in the same table.

```
ManagerID REFERENCES Employee (EmployeeID)
```

2. Composite Key

Definition:

A composite key is a primary key made up of **two or more columns** to uniquely identify a record.

Example:

```
PRIMARY KEY (OrderNo, ProductNo)
```

3. The LIKE operator with pattern matching

Definition:

The **LIKE** operator is used to search for a specified pattern in a column.

Example:

```
SELECT * FROM Client_Master  
  
WHERE City LIKE '_a%';
```

4. DROP TABLE command

Definition:

The **DROP TABLE** command permanently deletes a table along with its data and structure.

Example:

```
DROP TABLE Client_Master;
```

5. Full Outer Join

Definition:

A Full Outer Join returns all records from both tables, matching rows where available and NULL where no match exists.

Example:

```
SELECT *  
  
FROM Client_Master c  
  
FULL OUTER JOIN Sales_Order o  
  
ON c.ClientNo = o.ClientNo;
```

